Hierarchical porous carbon nanosheets derived from rice husk as coating layer for the anode of lead-carbon battery

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Highlights

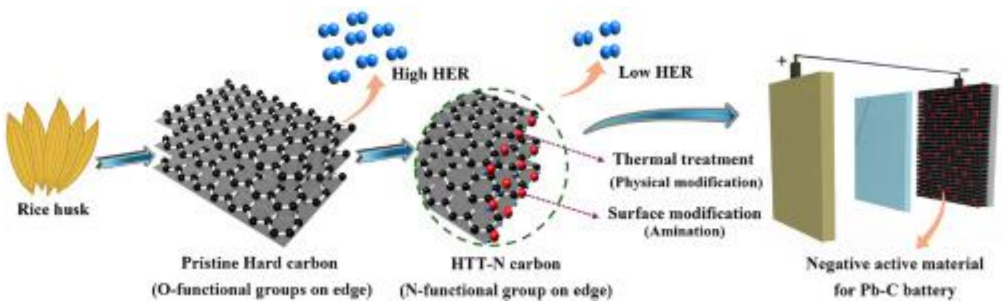
- Introduced dual modification for rice husk–derived activated carbon.
- Nitrogen functionalization enhances conductivity and active surface sites.
- HTT-N electrode achieved high discharge capacity of 1.89 Ah g^{-1}.
- HTT-N modified carbon reduces lead sulfate formation, extending battery lifespan.
- Comprehensive analysis confirms enhanced electrochemical and structural properties.

Abstract

Biomass-derived porous carbons offer a sustainable and cost-effective strategy to enhance the performance of lead–carbon batteries (LCBs), but challenges remain in controlling surface chemistry and suppressing the hydrogen evolution reaction (HER). In this study, we introduce a dual modification approach for rice husk–derived activated carbon (RHAC), involving thermal treatment at 1000 °C followed by nitrogen functionalization via amination, to produce a surface-engineered material designated as

HTT-N. Unlike conventional additive mixing, HTT-N is applied as a thin coating layer on the negative electrode to improve interfacial stability and redox efficiency. The HTT-N carbon retains a hierarchical porous structure and exhibits partial graphitic ordering, high electrical conductivity, and a nitrogen content of 2.08 wt%. These features collectively contribute to accelerated $Pb/PbSO_4$ redox kinetics, enhanced charge transport, and effective HER suppression. Electrochemical tests reveal a high specific discharge capacity of 1.89 Ah g^{-1} and outstanding cycling durability exceeding 40,000 cycles under high-rate partial-state-of-charge (HR-PSOC) conditions. These findings demonstrate that the synergistic effects of structural reconstruction and surface chemical tuning lead to a high-performance, metal-free, and scalable electrode material. The proposed HTT-N carbon coating thus represents a promising pathway for advancing next-generation LCB technologies using low-cost, biomass-derived resources.

Graphical abstract



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Introduction

The growing global transition toward renewable energy sources has increased the demand for dependable and cost-effective energy storage systems applicable to both mobile and stationary applications. Among commercial energy storage technologies, lead–acid batteries (LABs) continue to be competitive due to their low manufacturing cost, proven safety, well-developed recycling infrastructure, and scalability across diverse power requirements [1,2]. According to the International Lead Association (ILA), over 60% of LABs are utilized in automotive applications, while the rest support a wide spectrum of stationary uses, including uninterruptible power supplies (UPS), industrial backup, and grid-scale energy storage systems (ESS) [3]. The LAB market, estimated at USD 79.9 billion in 2021, is projected to reach USD 115.1 billion by 2030, highlighting its continued relevance despite the emergence of alternative technologies [4]. However, under high-rate partial-state-of-charge (HR-PSOC) operating conditions, conventional LABs experience significant limitations due to the accumulation of irreversible lead sulfate ($PbSO_4$) crystals on the negative electrode surface [5,6]. This phenomenon reduces charge acceptance and shortens battery life, motivating the development of lead–carbon batteries (LCBs). LCBs incorporate conductive carbon additives into the negative active material, which facilitate improved charge acceptance, alleviate sulfation, and prolong cycling durability [7,8]. Functionally, LCBs bridge the operational characteristics of both batteries and supercapacitors, exhibiting higher power density, faster charge–discharge capability, and enhanced long-term cycling stability—attributes particularly suited for hybrid automotive systems and renewable energy storage applications [9,10]. Despite these advances, an ongoing challenge in LCBs is the hydrogen evolution reaction (HER) that takes place at the negative electrode during charging. HER not only leads to electrolyte depletion and gas generation but also deteriorates the structural integrity of electrodes [11,12]. This undesirable side reaction is exacerbated by catalytically active edge sites prevalent in conventional activated carbon materials [13]. Several strategies have been proposed to suppress HER, such as doping with metal oxides (e.g., Bi_2O_3, Ga_2O_3, $SnSO_4$) or introducing heteroatoms like nitrogen or phosphorus into the carbon matrix [14,15]. However, many of these methods suffer from drawbacks including synthetic complexity, limited long-term stability, and insufficient chemical durability in acidic electrolytes [16]. Recent studies have demonstrated that the electrochemical behavior of carbon additives in LCBs is governed not only by surface area and porosity but also by the density of edge sites and microstructural stability [17,18]. In this context, biomass-derived activated carbon particularly that obtained from rice husks has attracted increasing attention. Rice husk-derived activated carbon (RHAC) exhibits a high specific surface area, well-developed hierarchical porosity, and favorable electrical conductivity [19,20]. RHAC has shown potential in enhancing $Pb/PbSO_4$ redox kinetics and facilitating uniform lead deposition. However, RHAC produced via conventional chemical activation (e.g., KOH treatment) often contains a high density of reactive edge sites (~8.5%), which catalyze HER and impair long-term electrochemical performance under HR-PSOC conditions [21].

In this study, we introduce a surface-engineered rice husk-derived activated carbon (RHAC) developed through a two-step post-treatment process consisting of thermal treatment (HTT) and subsequent nitrogen functionalization via amination. The thermal process restructures reactive edge sites by removing oxygen-containing functional groups, improving electrical conductivity and crystallinity. The following nitrogen doping passivates residual HER-active sites by introducing electron-rich amine groups. Distinct from conventional methods that blend carbon additives into the bulk of the negative active material [22,23], our surface modified RHAC was applied as a thin coating layer on the surface of the lead electrode [19,24,25]. This coating serves as an interfacial buffer between the electrode and electrolyte, functioning not only as a conductive network but also as a chemical barrier that suppresses hydrogen evolution, inhibits irreversible $PbSO_4$ buildup, and enhances interfacial redox kinetics. In contrast, bulk-mixed carbon additives mainly improved electrical conductivity and $PbSO_4$ dispersion within the electrode matrix but offer limited interfacial protection. Electrochemical results clearly demonstrate the superiority of the coating approach. The HTT-N coated electrode achieved a high specific discharge capacity of 1.89 Ah g^{-1} and excellent long-term cycling stability, exceeding 40,000 cycles under HR-PSOC conditions. These findings underscore the critical role of interfacial design and surface chemistry in optimizing lead–carbon battery performance and highlight the HTT-N material as a promising, sustainable, metal-free alternative for next-generation energy storage systems.

Section snippets

Materials

Rice husk was obtained from a local online market (Coupang, Republic of Korea). Potassium hydroxide (KOH), sulfuric acid (H_2SO_4) (Daejung Chemicals & Metals Co., Ltd., Korea, 70%), Diethylene triamine, anhydrous ethyl alcohol, commercial activated carbon (BS7, PCT9), sulfuric acid, sodium carboxymethyl cellulose (CMC, Sigma-Aldrich Corporation), styrene-butadiene rubber (SBR, 40 wt% dispersion in water, Hansol Chemical Co., Ltd.), an absorbent glass mat (AGM) (0.9 mm, Sebang Global Battery Co,

Structural, physicochemical, and surface chemical evolution of RHAC by thermal treatment and amination

To elucidate the effects of thermal treatment and chemical treatment on the structural and surface characteristics of RHAC, a comprehensive physicochemical analysis were conducted (Fig. 1.) The XRD patterns for all samples (Fig. 2(a)) revealed broad, featureless profiles without sharp crystalline peaks, indicating a lack of long-range graphitic stacking. A broad hump centered at $2\theta \approx 22–25°$, typically corresponding to the (002) diffraction plane of turbostratic or amorphous carbon, was observed

Conclusion

In this study, we proposed a novel and sustainable strategy to enhance the electrochemical performance of lead–carbon batteries by structurally and chemically modifying rice husk–derived activated carbon (RHAC). Through a dual post-treatment process—thermal annealing at 1000 °C followed by nitrogen functionalization via amination—we successfully tuned the structural order and surface chemistry of the carbon, resulting in the HTT-N material. Unlike conventional additive-type approaches that

CRediT authorship contribution statement

Priyadarshini Venkatachalam: Writing – original draft, Formal analysis, Data curation, Conceptualization. **Nagaraj Murugan:** Writing – review & editing, Formal analysis, Data curation. **Karmegam Dhanabalan:** Writing – review & editing, Formal analysis. **Min Kang:** Formal analysis. **Yu Rim Choi:** Formal analysis. **Seung Hwi Youn:** Data curation. **Seon Yeong Noh:** Formal analysis. **Seungki Hong:** Formal analysis, Data curation. **Hiroyuki Muramatsu:** Writing – review & editing, Formal analysis, Data curation..

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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